KAIZEN MATHS

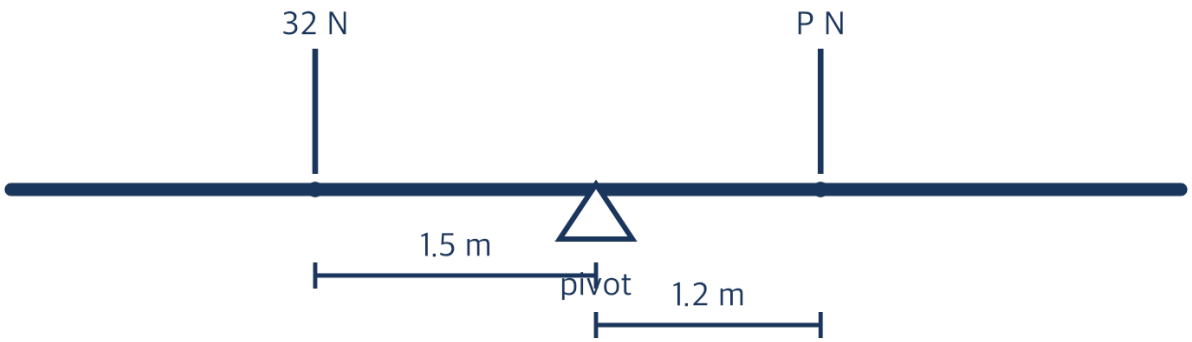
Lesson starter

Moments and Equilibrium

Principle of Moments

Name _____

SA1



The beam is in equilibrium. Find the value of P.

MEC-4D51-L2 · SA

Show your working

KAIZEN MATHS

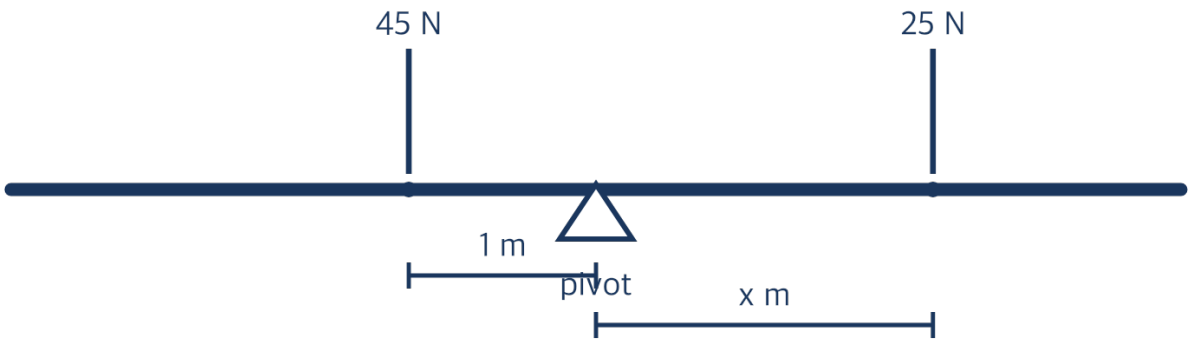
Lesson starter

Moments and Equilibrium

Principle of Moments

Name _____

SB1



The beam is in equilibrium. Find the distance x.

MEC-4D51-L2 · SB

Show your working

KAIZEN MATHS

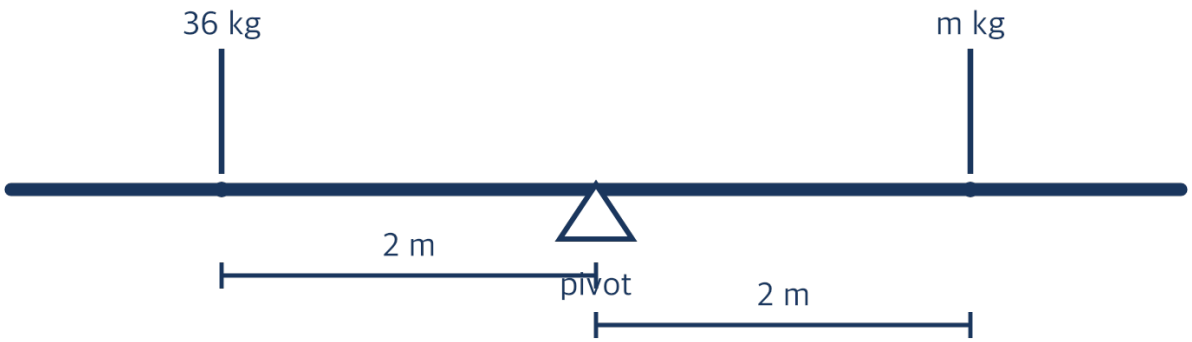
Lesson starter

Moments and Equilibrium

Principle of Moments

Name _____

SC1



Two children balance on a seesaw. Find the unknown mass m.

MEC-4D51-L2 · SC

Show your working

KAIZEN MATHS

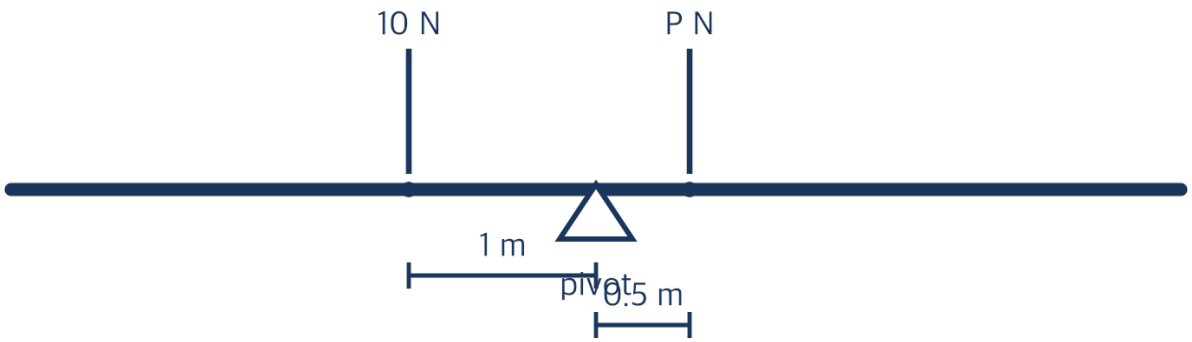
Lesson starter

Moments and Equilibrium

Principle of Moments

Name _____

SD1



The beam is in equilibrium. Find the value of P.

MEC-4D51-L2 · SD

Show your working

KAIZEN MATHS

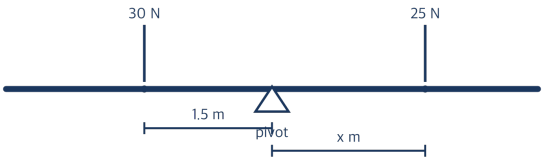
Exit ticket

Moments and Equilibrium

Principle of Moments

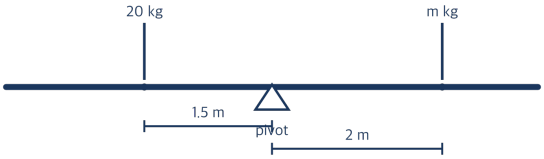
Name _____

EA1



The beam is in equilibrium. Find the distance x.

EA2



Two children balance on a seesaw. Find the unknown mass m.

Show your working

KAIZEN MATHS

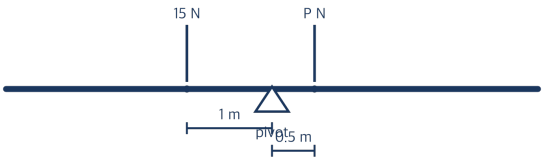
Exit ticket

Moments and Equilibrium

Principle of Moments

Name _____

EB1



The beam is in equilibrium. Find the value of P.

MEC-4D51-L2 · EB

Show your working

KAIZEN MATHS

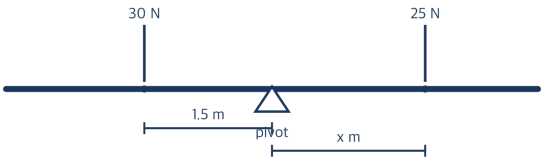
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Moments and Equilibrium

Principle of Moments

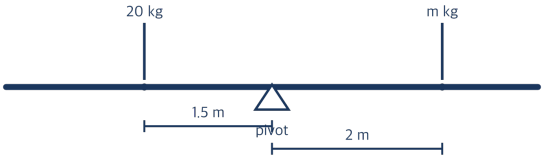
Name _____

EA1



The beam is in equilibrium. Find the distance x.

EA2



Two children balance on a seesaw. Find the unknown mass m.

MEC-4D51-L2 · EA

Show your working

KAIZEN MATHS

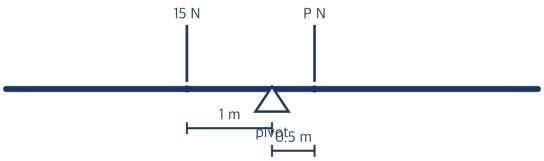
Exit ticket

Moments and Equilibrium

Principle of Moments

Name _____

EB1



The beam is in equilibrium. Find the value of P.

MEC-4D51-L2 · EB

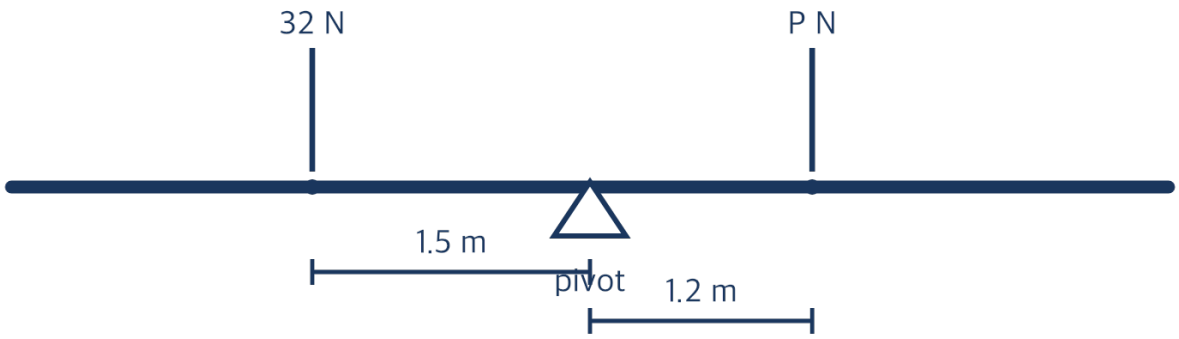
Show your working

Teacher answers & working

Moments and Equilibrium · Principle of Moments

MEC-4D51-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



The beam is in equilibrium. Find the value of P.

Answer:

$P = 40$ N

Working:

Since the beam is in equilibrium, clockwise moment equals anticlockwise moment.

clockwise moment = anticlockwise moment

Write the moment equation about the pivot.

$P \times 1.2 = 32 \times 1.5$

Solve for P.

$P \times 1.2 = 48$

$P = 48 \div 1.2 = 40$

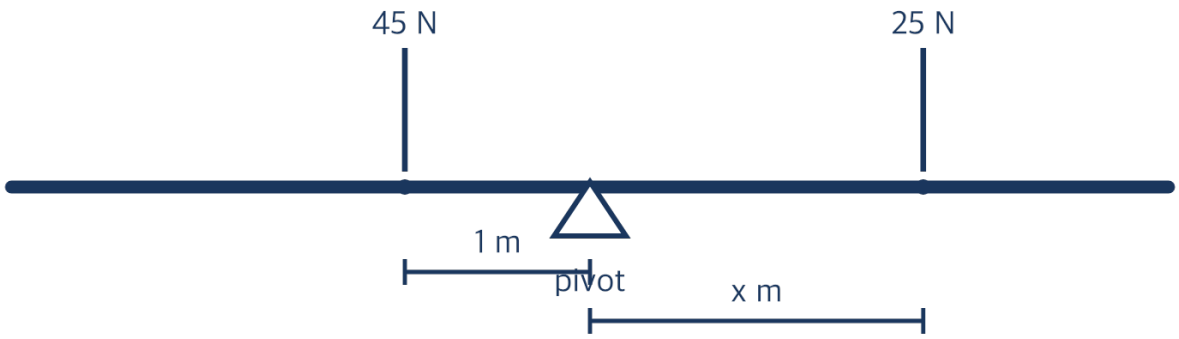
$P = 40$ N.

Teacher answers & working

Moments and Equilibrium · Principle of Moments

MEC-4D51-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



The beam is in equilibrium. Find the distance x.

Answer:

$x = 1.8$ m

Working:

Use the principle of moments.

clockwise moment = anticlockwise moment

Write the moment equation about the pivot.

$25 \times x = 45 \times 1$

Solve for x.

$25x = 45$

$x = 45 \div 25 = 1.8$

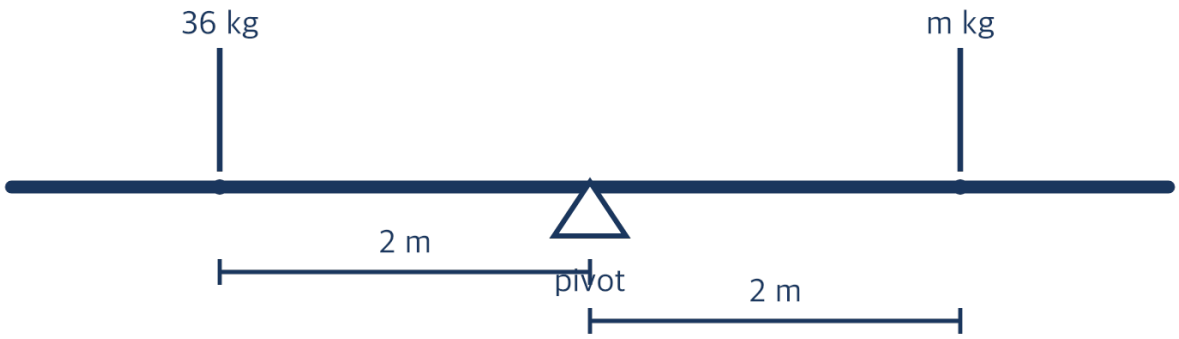
$x = 1.8$ m.

Teacher answers & working

Moments and Equilibrium · Principle of Moments

MEC-4D51-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Two children balance on a seesaw. Find the unknown mass m.

Answer:

$m = 36\ kg$

Working:

For masses on a seesaw, each weight is mg. The factor g appears on both sides and cancels.

$36g \times 2 = mg \times 2$

Cancel g and write the moment equation using masses.

$36 \times 2 = m \times 2$

Solve for m.

$72 = 2m$

$m = 72 \div 2 = 36$

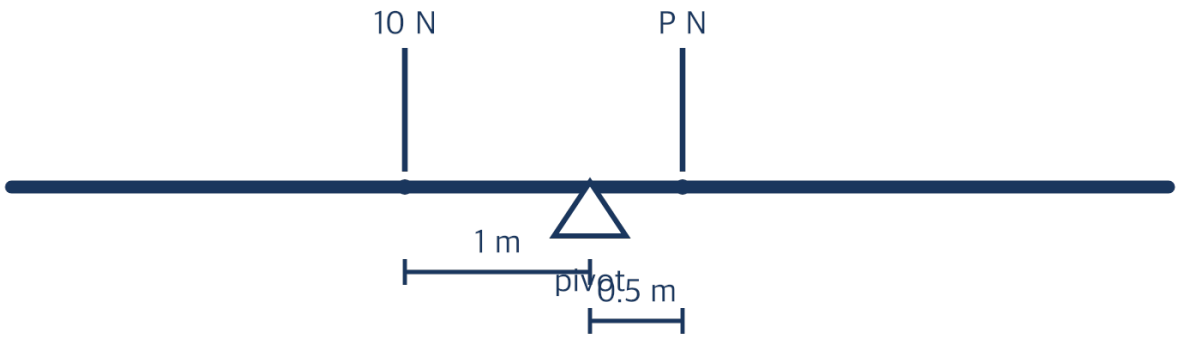
The unknown mass is 36 kg.

Teacher answers & working

Moments and Equilibrium · Principle of Moments

MEC-4D51-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



The beam is in equilibrium. Find the value of P.

Answer:

$P = 20$ N

Working:

Since the beam is in equilibrium, clockwise moment equals anticlockwise moment.

clockwise moment = anticlockwise moment

Write the moment equation about the pivot.

$P \times 0.5 = 10 \times 1$

Solve for P.

$P \times 0.5 = 10$

$P = 10 \div 0.5 = 20$

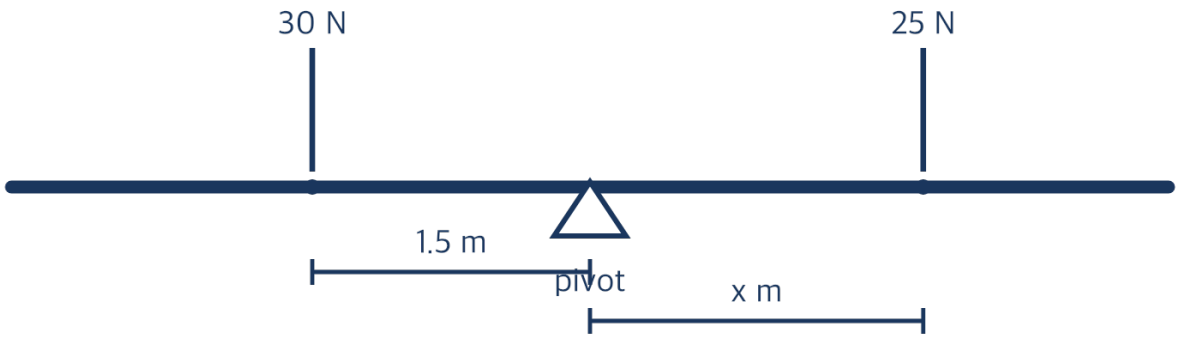
$P = 20$ N.

Teacher answers & working

Moments and Equilibrium · Principle of Moments

MEC-4D51-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



The beam is in equilibrium. Find the distance x.

Answer:

$x = 1.8$ m

Working:

Use the principle of moments.

clockwise moment = anticlockwise moment

Write the moment equation about the pivot.

$25 \times x = 30 \times 1.5$

Solve for x.

$25x = 45$

$x = 45 \div 25 = 1.8$

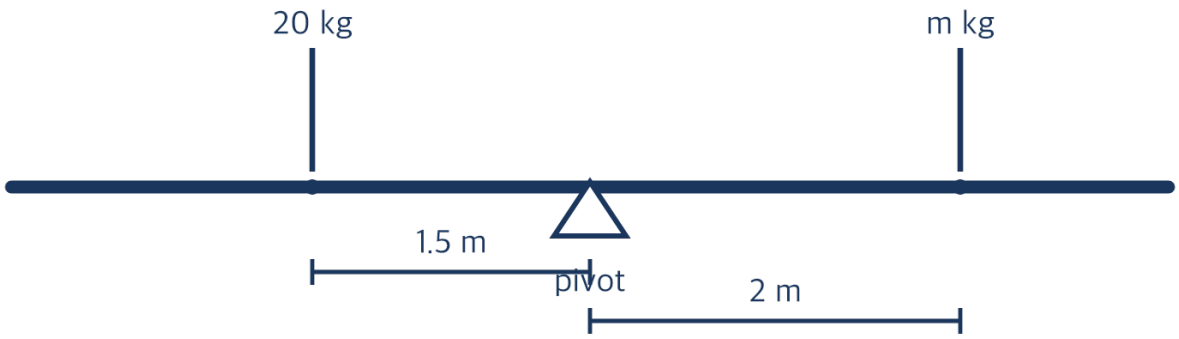
$x = 1.8$ m.

Teacher answers & working

Moments and Equilibrium · Principle of Moments

MEC-4D51-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA2



Two children balance on a seesaw. Find the unknown mass m.

Answer:

$m = 15\ kg$

Working:

For masses on a seesaw, each weight is mg. The factor g appears on both sides and cancels.

$20g \times 1.5 = mg \times 2$

Cancel g and write the moment equation using masses.

$20 \times 1.5 = m \times 2$

Solve for m.

$30 = 2m$

$m = 30 \div 2 = 15$

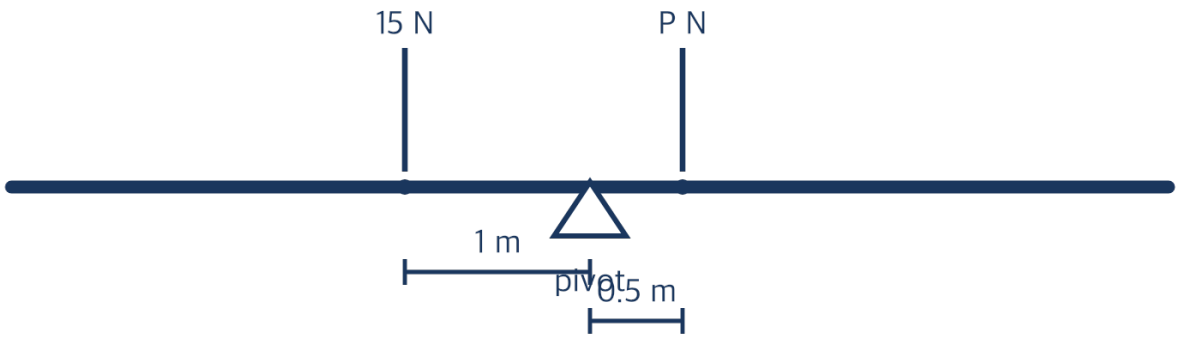
The unknown mass is 15 kg.

Teacher answers & working

Moments and Equilibrium · Principle of Moments

MEC-4D51-L2 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



The beam is in equilibrium. Find the value of P.

Answer:

$P = 30$ N

Working:

Since the beam is in equilibrium, clockwise moment equals anticlockwise moment.

clockwise moment = anticlockwise moment

Write the moment equation about the pivot.

$P \times 0.5 = 15 \times 1$

Solve for P.

$P \times 0.5 = 15$

$P = 15 \div 0.5 = 30$

$P = 30$ N.